**UNIVERSIDAD SAN FRANCISCO DE QUITO**
**COLEGIO: CIENCIAS E INGENIERÍAS**
**Semestre:** 202520 - Segundo Semestre 2025/2026

# Phase 2 Project Specification

*Simulated Bitcoin Exchange*

### How Phase 2 will be graded

This phase will be also graded by running your published Docker images directly from Docker Hub. Your code will not be rebuilt locally during grading. All published images must run on both ARM and AMD CPUs.

You must still upload the full source code to D2L. Include a README.txt file with the submission. The README.txt must list all team members exactly as they appear in D2L, describe each person's contribution, and provide complete instructions for running this phase of the project.

Each phase builds on the previous one. The project consists of three phases, and the third phase is the final project.

## 1. Big picture

Phase 1 established the mechanisms for bringing BTC into the exchange through mining and on-chain deposits. Phase 2 should make the platform fully usable by enabling clients to trade BTC and U.S. dollars with one another at different prices. In this stage, the exchange must track internal client balances, accept orders, match buyers and sellers, execute trades, apply the owner’s fee, and display the results clearly in the interface.

This phase assumes that some miners are already clients of the exchange and have deposited BTC into the exchange wallet in at least the threshold amount defined in Phase 1. It also assumes that 30 additional clients have exchange accounts funded with an initial balance of USD 30,000 each. At this stage, all exchange clients should be able to execute trades.

All trades are internal to the exchange's internal accounts. In this phase, the owner's 2% fee is calculated for each trade and added to an internal BTC balance belonging to the owner. Blockchain transactions involving this fee are not yet included in this phase.

## 2. Exchange-activity related definitions

**Order**: a request to trade.

**Trade**: the actual exchange that happens when orders match.

**Buy:** means buying BTC.

**Sell:** means selling BTC.

## 3. System roles in Stage 2

| Role | Meaning in Stage 2 |
|---|---|
| Exchange | The backend and database that manage internal balances, orders, matches, trades, and reporting. |
| Client | A user account inside the exchange with internal BTC and USD balances. |
| Seller | A client offering BTC for sale inside the exchange. |
| Buyer | A client using USD to buy BTC inside the exchange. |
| Owner | The exchange owner who receives a 2% fee on each trade, paid in BTC. |
| Exchange wallet | The blockchain wallet controlled by the exchange. |

## 4. Balance concepts

| Balance idea | Meaning |
|---|---|
| Available BTC | BTC that the client can still use for a new sell order or, later, a withdrawal. |
| Reserved BTC | BTC that has been locked because the client already placed a sell order using it. |
| Available USD | USD that the client can still use for a new buy order. |
| Reserved USD | USD that has been locked because the client already placed a buy order using it. |

| | |
|---|---|
| Exchange-controlled BTC | Real BTC already received and verified by the exchange on the blockchain. |
| Internal BTC | The BTC balance shown to a client inside the exchange database. This is the amount the client can trade inside the market. |

## 5. Order types

| Order type | Requirement | Meaning |
|---|---|---|
| Limit buy | Required | A client wants to buy a given amount of BTC at or below a chosen price. |
| Limit sell | Required | A client wants to sell a given amount of BTC at or above a chosen price. |
| Market order | Optional but recommended | A client wants immediate execution against the best available opposite-side orders. |

## 6. Exchange rules for buy and sell requests

**For sell orders:**

- A lower sell price has priority
- For the same price level, earlier order first

**For buy orders:**

- A higher buy price has priority
- For two buy orders at the same price, the one entered earlier in time goes first

**An order is either fully matched or not matched**

## 7. Trade execution and owner fee

Each successful match produces a trade. A trade is the record of one executed amount between one buy order and one sell order. The trade should update balances immediately inside the exchange database.

### Required fee rule

The owner will receive a 2% fee on every trade, paid in BTC. For this stage, the clearest implementation is to deduct the fee from the BTC side of the trade. That means the buyer receives slightly less BTC than the raw traded amount, and the deducted BTC is credited to the owner fee account.

**For example:**

| Item | Example value |
|---|---|
| Executed trade amount | 1.00000000 BTC |
| Owner fee (2%) | 0.02000000 BTC |
| BTC credited to buyer | 0.98000000 BTC |
| BTC credited to Exchange owner | 0.02000000 BTC |
| USD debited from buyer | Trade price × 1.00000000 BTC |
| USD credited to seller | Trade price × 1.00000000 BTC |

## 8. Required user interface elements

A client list or client selector: The ability to pick **one or more clients** and be able to see the status of their **accounts**:

- Balances of BTC and US dollars
- Whether they have open trades.

An open orders table.

The last completed orders table; viewable until the user deletes it.

The exchange owner total funds accrued from trade fees.

## 9. Validations

- Reject an order if the client does not have enough available funds.
- Reject negative or zero price values.
- Reject negative or zero amount values.
- Prevent the same funds from being used in two open orders at the same time.